

Spiking Neural Networks (SNN)

NECKER

September 6, 2026

Contents

1	Introduction: The Neuromorphic Paradigm	2
2	The Biological Neuron and Information Encoding	2
2.1	Signal Encoding Mechanisms (Spike Coding)	2
2.1.1	1. Rate Coding	2
2.1.2	2. Temporal Coding	3
2.1.3	3. Population Coding	3
3	Mathematical Models of the Spiking Neuron	5
3.1	The Leaky Integrate-and-Fire (LIF) Model	5
3.1.1	Biophysical Explanation of the Synaptic Transmission Chain	5
3.1.2	Analytical Derivation via Kirchhoff's Law	8
3.2	Advanced and Conductance-Based Models	11
4	Synapses and Synaptic Current Models	12
4.1	Delta-Current Model:	12
4.2	Conductance-Based (Exponential) Model:	12
5	Learning in Neuromorphic Networks	13
5.1	Biological Foundations: What Does "Learning" Mean Physically?	13
5.2	1. Unsupervised Learning: Hebbian Plasticity and STDP	14
5.2.1	Mathematical Training Rule:	15
5.3	2. Supervision via Surrogate Gradients	16
5.4	3. ANN-to-SNN Conversion	17
6	SNNs in Embodied AI: Advantages and Neuromorphic Hardware	19
7	Conclusions and Research Frontiers	19

1 Introduction: The Neuromorphic Paradigm

In the field of Artificial Intelligence, while computational attention is currently dominated by Transformer-based architectures and Large Language Models, there is a line of research focused not on semantic depth, but on energy efficiency, biological latency, and physiological realism: **Spiking Neural Networks (SNNs)**.

SNNs represent the "third generation" of neural networks, succeeding first-generation perceptrons (binary outputs) and second-generation ANNs (continuous activation functions such as Sigmoid or ReLU). The fundamental distinction lies in the treatment of time: in ANNs, time is discretized per layer and activations are frame-based; in SNNs, time is an intrinsic continuous variable and information is conveyed by the exact instant of emission of an action potential, or **spike**.

This paradigm shift holds revolutionary promises for Embodied AI, especially when paired with neuromorphic sensors (such as Event Cameras) and dedicated hardware (such as Intel Loihi or IBM TrueNorth chips), promising energy consumption in the milliwatt range for complex visuo-motor tasks.

2 The Biological Neuron and Information Encoding

To understand SNNs, it is necessary to examine a biological neuron, which receives chemical signals through dendrites that alter the electrical potential of its cytoplasmic membrane. If this potential exceeds a certain critical threshold, the neuron "fires" a stereotypical electrical pulse (spike) along the axon, propagating to the synapses of subsequent neurons.

2.1 Signal Encoding Mechanisms (Spike Coding)

In conventional neural networks (ANNs), information is represented by continuous 32-bit floating-point values (e.g., a neuron's activation is $x = 0.85$). Conversely, in a Spiking Neural Network (SNN), all action potentials (spikes) possess stereotypical amplitude and duration: the instantaneous output of a neuron is a purely binary event $S(t) \in \{0, 1\}$.

Consequently, a fundamental engineering question arises: **how do we translate a continuous physical quantity $x \in [0, 1]$ into a discrete temporal sequence of spikes?**

In neuroscientific and engineering literature, there is no single answer; three alternative paradigms (or transmission protocols) have been formalized, each based on different computational assumptions.

2.1.1 1. Rate Coding

In this paradigm, information resides exclusively in the **total number of spikes** emitted within a fixed time window of duration T , regardless of their precise temporal distribution. The stimulus intensity x is linearly mapped to the average firing rate (r):

$$r = \frac{1}{T} \int_0^T S(t) dt \approx \frac{N_{\text{spike}}}{N_{\text{step}}} \propto x$$

$\propto$ indicates direct proportionality between two quantities.

Practical Example: Consider a pressure sensor on a gripper measuring intense contact ($x = 0.80$) or slight contact ($x = 0.20$) over a discretized interval of $T = 10$ time steps:

- **Strong pressure** ($x = 0.80$): The neuron generates 8 spikes over 10 intervals:

$$S(t) = [1, 1, 0, 1, 1, 1, 0, 1, 1, 1] \implies \frac{8}{10} = 0.80$$

- **Light pressure** ($x = 0.20$): The neuron generates only 2 spikes over 10 intervals:

$$S(t) = [0, 0, 1, 0, 0, 0, 0, 1, 0, 0] \implies \frac{2}{10} = 0.20$$

Trade-off: It is the most robust method against sensory noise (the loss of a single pulse only marginally alters the average firing rate) and is the foundation for direct ANN-to-SNN conversion. However, it is **highly inefficient in terms of latency and power consumption**: the receiving neuron must wait for the entire window T to accumulate and estimate the correct rate.

2.1.2 2. Temporal Coding

Unlike Rate Coding, here the exact time of occurrence of each individual spike carries critical information. One does not count the quantity of pulses, but rather "reads the clock" at the instant of arrival. There are two primary implementations:

- **Time-to-First-Spike (TTFS or Latency):** The stimulus intensity is inversely proportional to the latency of the first spike relative to the onset of detection ($t = 0$). A strong stimulus drives the membrane potential to threshold much faster than a weak stimulus:

$$t_{\text{spike}} = t_0 - \tau \ln(x)$$

Example: For strong pressure $x = 0.80$, the neuron emits a single spike almost instantaneously at step $t = 1$ ms; for light pressure $x = 0.20$, the neuron takes time to integrate charge and emits the spike at step $t = 9$ ms. The postsynaptic neuron determines the value by measuring the elapsed time from stimulus onset, achieving nearly instantaneous decision latency with a single spike.

- **Inter-Spike Interval (ISI):** Information is encoded in the relative temporal distance $\Delta t = t_{k+1} - t_k$ between two consecutive pulses, where a compressed interval corresponds to a strong stimulus and an extended interval to a weak stimulus. Unlike TTFS, which resolves reading with a single event but requires a global synchronization signal ($t = 0$), ISI requires a continuous sequence of at least two spikes to define the interval, but operates asynchronously and self-sufficiently without external temporal references.

Trade-off: Maximum energy efficiency (often requiring only a single spike per event) and millisecond reactivity, ideal for closed-loop reactive control. Conversely, it is extremely vulnerable to temporal noise (*jitter*): even minimal oscillation in the hardware clock distorts the transmitted datum.

2.1.3 3. Population Coding

Because individual biological neurons are subject to significant stochastic components, the nervous system often distributes the encoding of a single continuous variable across a collection (*pool*) of cooperating neurons. Each neuron in the population possesses a Gaussian "tuning curve" centered on a specific portion of the domain:

$$f_i(x) = \exp\left(-\frac{(x - \mu_i)^2}{2\sigma_i^2}\right)$$

- $f_i(x) \in [0, 1]$: Represents the **activation level** (or normalized response) of the i -th neuron in the population in response to the scalar stimulus x . It reaches a maximum value of 1 when the stimulus matches the neuron's preference exactly, and decays asymptotically toward 0 as the stimulus moves away.
- x : The **continuous stimulus value** at the input (e.g., contact force measured by a sensor or joint angle) that the neuronal population must encode.
- μ_i : The **tuning center** (or mean) of the i -th neuron. It defines the specific stimulus value to which the neuron is maximally sensitive: for $x = \mu_i$, the exponent vanishes and $f_i(\mu_i) = \exp(0) = 1$. (Even though the result of the numerator is zero, the exponential brings the output to 1, so this value represents the exact "class" of the neuron.)
- $(x - \mu_i)^2$: Represents the **squared Euclidean distance** between the current stimulus x and the neuron's preferred value μ_i . The square ensures that the response decays symmetrically in both directions (both above and below).
- σ_i : The **tuning width** (or standard deviation) of the receptive field. It controls the neuron's selectivity: a small σ_i produces a narrow, highly specific response, whereas a large σ_i yields a broad, diffuse response, increasing overlap with adjacent neurons.
- $2\sigma_i^2$: The **scaling factor in the denominator** that normalizes the spread of the Gaussian bell curve. As this term increases, the decay slope flattens.
- $\exp(-\cdot)$: The negative exponential function maps the normalized distance into the interval $(0, 1]$, transforming metric discrepancy into a smooth, differentiable bell-shaped profile (*Radial Basis Function*, RBF).

Once the various activation values $f_i(x) \in [0, 1]$ are computed, the initial input value can be reconstructed using the formula:

$$\hat{x} = \frac{\sum_i f_i \cdot \mu_i}{\sum_i f_i}$$

Practical Example: Let us fix the dimensional parameters for a population of three neurons (N_1, N_2, N_3) dedicated to encoding the wrist rotation angle $x \in [0^\circ, 180^\circ]$:

- Tuning centers: $\mu_1 = 0^\circ, \mu_2 = 90^\circ, \mu_3 = 180^\circ$.
- Constant tuning width: $\sigma_1 = \sigma_2 = \sigma_3 = 30^\circ$.
- Denominator scaling factor: $2\sigma^2 = 2 \cdot (30)^2 = 2 \cdot 900 = 1800$.

Suppose the sensor detects an actual angle of $x = 120^\circ$. Let us trace the calculation step by step for each of the three neurons:

1. **Neuron N_1 (Specialized at $\mu_1 = 0^\circ$):**

- Squared distance: $(x - \mu_1)^2 = (120 - 0)^2 = 120^2 = 14400$.
- Normalized argument: $\frac{(x - \mu_1)^2}{2\sigma_1^2} = \frac{14400}{1800} = 8.0$.
- Activation level:

$$f_1(120^\circ) = \exp(-8.0) \approx 0.0003 \approx 0.03\%$$

The significant discrepancy between x and μ_1 almost entirely extinguishes the firing of the first neuron.

2. **Neuron N_2 (Specialized at $\mu_2 = 90^\circ$):**

- Squared distance: $(x - \mu_2)^2 = (120 - 90)^2 = 30^2 = 900$.
- Normalized argument: $\frac{(x - \mu_2)^2}{2\sigma_2^2} = \frac{900}{1800} = 0.5$.
- Activation level:

$$f_2(120^\circ) = \exp(-0.5) \approx 0.6065 \approx 60.65\%$$

The deviation falls within one standard deviation ($|120 - 90| = \sigma$), yielding a robust, dominant activation.

3. **Neuron N_3 (Specialized at $\mu_3 = 180^\circ$):**

- Squared distance: $(x - \mu_3)^2 = (120 - 180)^2 = (-60)^2 = 3600$.
- Normalized argument: $\frac{(x - \mu_3)^2}{2\sigma_3^2} = \frac{3600}{1800} = 2.0$.
- Activation level:

$$f_3(120^\circ) = \exp(-2.0) \approx 0.1353 \approx 13.53\%$$

The distance of two standard deviations significantly reduces the response, leaving nonetheless a measurable residual component.

Value Reconstruction via Population Vector: The vector of normalized responses emitted by the sensory layer is:

$$\mathbf{f}(120^\circ) = [f_1, f_2, f_3]^T \approx [0.0003, 0.6065, 0.1353]^T$$

The postsynaptic circuit can thus reconstruct the original continuous stimulus by computing the weighted centroid of population activity (*Population Vector Decoding*):

$$\hat{x} = \frac{\sum_{i=1}^3 f_i(x) \cdot \mu_i}{\sum_{i=1}^3 f_i(x)} = \frac{(0.0003 \cdot 0^\circ) + (0.6065 \cdot 90^\circ) + (0.1353 \cdot 180^\circ)}{0.0003 + 0.6065 + 0.1353}$$

$$\hat{x} = \frac{0 + 54.585 + 24.354}{0.7421} = \frac{78.939}{0.7421} \approx 106.37^\circ$$

This residual deviation from 120° numerically illustrates the constraint of biological discretization: as the density of neurons in the population increases (e.g., adding intermediate centers μ every 15°), the estimate $\hat{x}$ converges asymptotically to the true value x .

Trade-off: Offers maximum resilience against physical faults: even in the presence of neuronal death or silence from an anomalous node, the population retains the overall metric estimate. The disadvantage lies in structural overhead: it requires a noticeably higher number of synaptic channels.

Paradigm	Key Mechanism	Spike Count	Latency	Primary Characteristic
Rate Coding	Average firing rate over window ΔT	High	High	Noise robustness; baseline for ANN-SNN conversion.
Temporal (TTFS)	Exact time instant of first pulse	Minimal (≈ 1)	Minimal	Near-zero power; vulnerable to hardware jitter.
Population	Activation vector of distributed pool	Medium	Low	Biological redundancy; resilience to single-node failures.

Table 1: Synoptic comparison of signal encoding protocols in neuromorphic architectures.

3 Mathematical Models of the Spiking Neuron

While we previously explored communication interpretation paradigms between neurons, we now need to understand how neurons themselves decide whether or not to activate. To do so, we examine the mathematical abstraction of the neuron. Various models exist, though the most widely used is **LIF**:

3.1 The Leaky Integrate-and-Fire (LIF) Model

The LIF model is the "workhorse" of modern SNNs for AI. It can be described as a parallel RC (resistor-capacitor) circuit.

3.1.1 Biophysical Explanation of the Synaptic Transmission Chain

To clarify the abstraction operated by SNN computational models relative to biological physiology, it is essential to trace the complete bioelectrochemical cycle governing unidirectional communication between a transmitting neuron (Neuron A, presynaptic) and a receiving neuron (Neuron B, postsynaptic), bathed in the extracellular matrix.

Physical and Molecular Components of the Biological Neuron

Before analyzing the temporal dynamics of the spike and synaptic transmission, it is helpful to inventory the structural and molecular components that form the bioelectrochemical "hardware" of a neuron:

- **Phospholipid Bilayer (The Membrane):** A lipid barrier a few nanometers thick physically separating internal cytoplasm from extracellular fluid. Impermeable to ions, it acts as a dielectric, turning the membrane into a **microscopic capacitor** (C) capable of storing electrical charge.
- **Passive or Leak Ion Channels (*Leak Channels*):** Continuously open protein pores (primarily selective for potassium K^+) that allow a restrained yet steady flow of ions across the wall, acting physically as a **leak resistor** (R).
- **The Sodium-Potassium Pump (Na^+/K^+ -ATPase):** An active molecular engine consuming metabolic energy (ATP) to continually expel 3 Na^+ ions and pump in 2 K^+ ions. It maintains chemical concentration gradients and sets the baseline energy level or **resting potential** ($V_{rest} \approx -70$ mV).
- **Voltage-Gated Channels (Na^+ and K^+):** Smart ion gates distributed along the soma and axon. They open or close automatically in response to local variations in membrane voltage, acting as non-linear switches responsible for rapid spike generation.
- **Voltage-Gated Calcium Channels (VGCCs):** Channels located exclusively on the axon terminal tip. They activate upon arrival of the electrical voltage wave, permitting entry of Ca^{2+} ions that act as a chemical trigger.
- **Synaptic Vesicles and Neurotransmitters (e.g., Glutamate):** Tiny lipid sacs clustered in the presynaptic terminal, filled with chemical messenger molecules ready to be released into the synaptic cleft.

- **Postsynaptic Receptors (AMPA/NMDA):** Receptor proteins located on the dendrites of the receiving neuron. They recognize and bind released glutamate, opening gates for an influx of Na^+ ions and translating the chemical signal into the input current $I_{\text{syn}}(t)$.

1. Impulse Genesis and Propagation in Neuron A

The dynamics begin in the cell body (soma) of Neuron A:

1. **Integration and Critical Threshold:** Depolarizing ionic currents drive the local membrane potential toward the activation threshold (carrying Na^+ ions):

$$V_m^A(t) \geq V_{\text{th}} \approx -50 \text{ mV}$$

2. **Action Potential (Polarity Inversion):** Exceeding this threshold instantaneously unlocks voltage-gated sodium (Na^+) channels. The massive influx of Na^+ abruptly inverts membrane polarity up to approximately +30 mV.
3. **Reset and Restoration of A's Membrane:** Reaching the positive peak, sodium channels inactivate and fast potassium (K^+) channels open, expelling K^+ ions into the local extracellular environment. This outflow of positive charge causes the potential to plummet back toward the baseline value:

$$V_m^A(t^+) = V_{\text{reset}} \approx -70 \text{ mV}$$

Meanwhile, the initial voltage reversal generates a depolarizing electrical wave propagating along the axon to the presynaptic terminal.

2. Electrochemical Transduction via Calcium (Ca^{2+}) and Vesicular Release

When the electrical wavefront at +30 mV reaches Neuron A's axon terminal, the conversion interface between electrical signal and chemical modulation activates:

- **Opening of Ca^{2+} Channels:** Terminal depolarization opens voltage-gated calcium channels (VGCCs).
- **Trigger Role of Calcium:** Given the high concentration of calcium outside the cell (approximately 10^4 times higher than in the cytoplasm), Ca^{2+} ions rush into the synaptic bouton. Calcium acts as a biomechanical catalyst: by binding to specific protein sensors (synaptotagmin and the SNARE complex), it triggers the fusion of synaptic vesicles with the outer membrane (*exocytosis*).
- **Neurotransmitter Emission:** Vesicles discharge discrete packets of excitatory neurotransmitter (typically **glutamate**) into the synaptic cleft, an intercellular gap of approximately 20 nm.

3. Reception and Current Generation in Neuron B (I_{syn})

Glutamate molecules diffuse across the synaptic cleft and bind to postsynaptic receptors (AMPA and NMDA) located on Neuron B's dendrites:

- Receptor-ligand binding causes the selective opening of ion channels, allowing a direct influx of Na^+ (and partially Ca^{2+}) from the extracellular fluid into Neuron B.
- This influx of positive charges constitutes the **synaptic current** $I_{\text{syn}}(t)$.
- Neuron B's membrane integrates this charge over time according to its membrane time constant τ_m :

$$\tau_m \frac{dV_m^B(t)}{dt} = -[V_m^B(t) - V_{\text{rest}}] + R \cdot I_{\text{syn}}(t)$$

If the current input $I_{\text{syn}}(t)$ (produced by one or more synchronous spikes) is intense enough to overcome resistive losses (*leak* mediated by K^+ efflux), potential $V_m^B(t)$ will also reach the critical threshold V_{th} , propagating the signal to subsequent network nodes.

4. Cycle Closure: Neurotransmitter Recycling and Calcium Extrusion

For the synapse to maintain temporal resolution and avoid saturation or excitotoxicity, the transmission environment is cleared within a few milliseconds:

1. **Intracellular Ca^{2+} Clearance:** Calcium ions inside A's terminal are rapidly sequestered into the endoplasmic reticulum and pumped out via active ATP-dependent pumps (PMCA) and sodium-calcium exchangers (NCX). Halting the Ca^{2+} influx immediately stops the exocytosis of further vesicles.
2. **Glutamate Reuptake:** Glutamate in the cleft does not decay passively; it is actively removed via dedicated membrane transporters (EAATs, *Excitatory Amino Acid Transporters*) located on both the presynaptic membrane and surrounding glial cells (astrocytes).
3. **Glutamate-Glutamine Cycle and Vesicle Endocytosis:**
 - In astrocytes, absorbed glutamate is enzymatically converted into **glutamine** (an inert molecule devoid of neurotransmitter activity), shuttled back to the presynaptic neuron, and enzymatically reconverted into glutamate (via glutaminase).
 - Concurrently, A's axon terminal membrane recovers lipid components from fused vesicles via *endocytosis*.
 - Vesicular transporters (VGLUTs) exploit a proton gradient to actively repackage recycled glutamate into new vesicles, restoring the pool of quanta for the next discharge.

Mapping to the SNN Computational Model: In Spiking Neural Networks for Embodied AI, simulating every single chemical reaction (local Ca^{2+} influx, mechanical vesicle fusion, free glutamate diffusion, and astrocyte-mediated recycling) would introduce an unsustainable computational burden for real-time robotic control. Therefore, the entire biophysical chain is summarized into a compact mathematical formulation:

$$I_{\text{syn}}^B(t) = \sum_j w_{jB} \sum_k \kappa(t - t_j^k)$$

This expression decomposes the effect of each incident pulse into three distinct operational elements:

- **Presynaptic spike trains (t_j^k):** Index j ranges over all upstream neurons connected to Neuron B, while t_j^k represents the exact moment (k -th time instant) at which neuron j emitted a binary spike.
- **Scalar synaptic weight (w_{jB}):** Summarizes into a single learnable parameter the overall "strength" of the communication channel between node j and node B. Biologically, it simultaneously condenses:
 1. Presynaptic efficiency (how many glutamate molecules are released per spike);
 2. Postsynaptic sensitivity (how many receptors and sodium channels open on the receiving membrane).

A high w_{jB} value means a single spike will cause a substantial influx of positive charge, whereas a value near zero renders the signal transmission nearly negligible.

- **Temporal response kernel ($\kappa(t - t_j^k)$):** When neurotransmitter binds to receptors, ion channels do not open and close instantaneously; they exhibit smooth dynamics with a rapid rise phase and an exponential decay tail. Function $\kappa(\Delta t)$ models this transient response: it quantifies how much current continues to flow Δt milliseconds after the spike arrives.

In the simplest and most efficient case for parallel computing on GPUs/NPUs (instantaneous current or delta model), this kernel is approximated by a Dirac delta function:

$$\kappa(t - t_j^k) = \delta(t - t_j^k)$$

In this simplified configuration, each spike emitted at time t immediately injects a discrete quantity of charge proportional to w_{jB} , delegating temporal persistence and discharge memory entirely to the membrane decay constant τ_m of the receiving neuron.

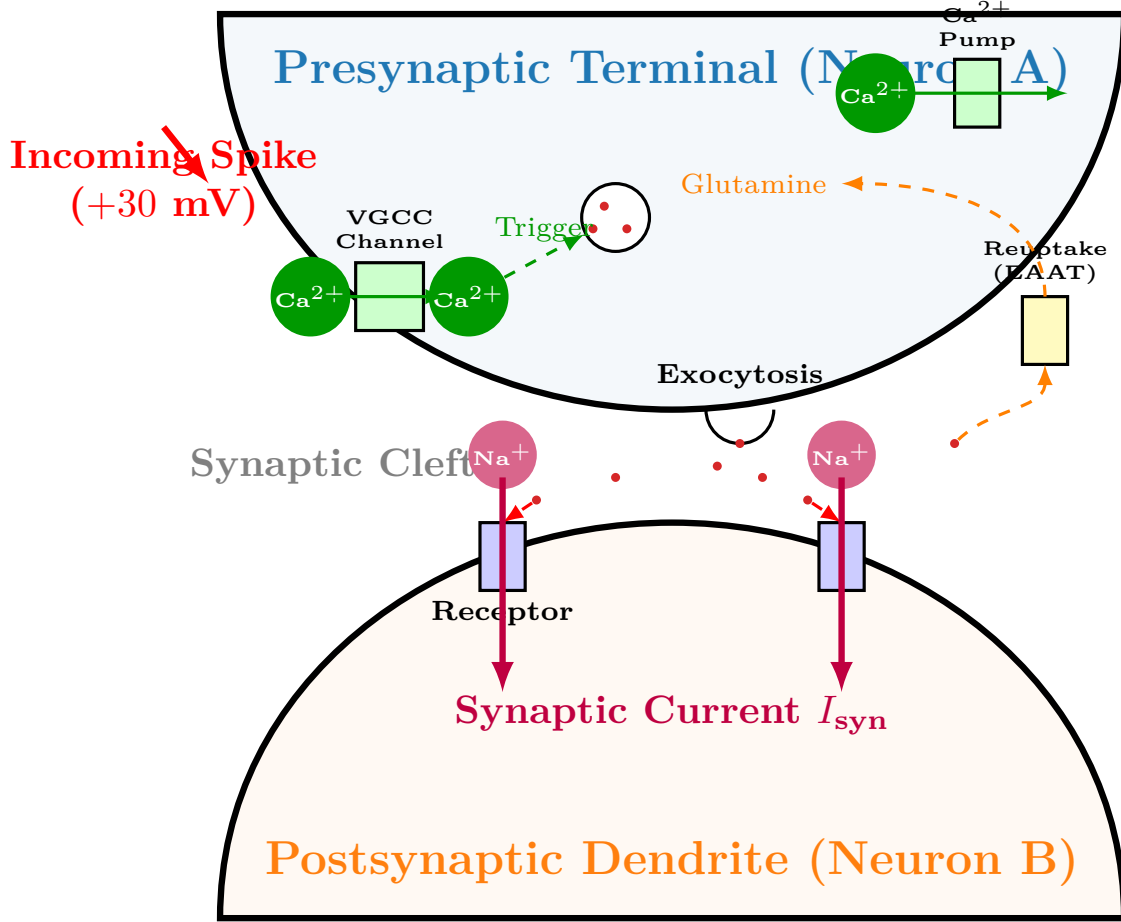


Figure 1: Detail of the bioelectrochemical transduction mechanism (Phases 2-5). The action potential's arrival unlocks VGCC channels, allowing Ca^{2+} entry. The latter acts as a mechanical switch, driving vesicles to release Glutamate into the cleft (exocytosis). The neurotransmitter binds to postsynaptic receptors, opening ion channels that generate the synaptic current I_{syn} as a flow of Na^+ into Neuron B. Concurrently, reuptake mechanisms and ion extrusion clear the environment for the next cycle.

3.1.2 Analytical Derivation via Kirchhoff's Law

To understand the physical origin of the differential equation of the LIF neuron, Kirchhoff's current law (nodal rule) is applied to the cell membrane, modeled as the separation boundary between internal cytoplasm and extracellular fluid.

At every instant t , electrical charge cannot accumulate indefinitely at an ideal point without producing voltage. Therefore, the total synaptic current $I_{\text{syn}}(t)$ entering the cell (carried by synaptic channel opening following incoming spikes) must split into only two possible physical paths:

$$I_{\text{syn}}(t) = I_C(t) + I_R(t)$$

Let us analyze the physical meaning and biological role of these two contributions:

- **Capacitive Accumulation Current ($I_C(t)$) — The Neuron's Short-Term Memory:** The phospholipid membrane is an extremely thin insulator (3–4 nm) preventing ions from crossing freely, behaving as an electrical capacitor of capacitance C . Because the charge stored on the plates is $Q(t) = C \cdot V_m(t)$ (a potential that changes over time), the capacitive current corresponds to the rate at which net charge deposits onto membrane surfaces over time:

$$I_C(t) = \frac{dQ(t)}{dt} = C \frac{dV_m(t)}{dt}$$

Physical meaning: It is the flow of positive ions (mainly Na^+ and Ca^{2+}) entering and accumulating along the inner face of the membrane. The stronger this flux, the faster the variation of cell voltage $\frac{dV_m(t)}{dt}$. It is this term that grants the neuron the ability to integrate multiple distinct inputs across time.

- **Resistive Leakage Current ($I_R(t)$) — Loss (*Leak*) via Passive Dissipation:** The membrane is not an ideal insulator: it contains passive ion channels (known as leak channels, selective for potassium K^+) that remain persistently open. Ohm's law applied across this barrier establishes that the current through these pores is driven not by absolute voltage, but by the electrochemical gradient relative to the natural equilibrium of the cell ($V_{\text{rest}} \approx -70$ mV):

$$I_R(t) = \frac{V_m(t) - V_{\text{rest}}}{R}$$

where R is the total ohmic resistance offered by the membrane.

Physical meaning: If the neuron has been excited and the potential rises above V_{rest} (for instance to -60 mV), the potential difference pushes potassium ions K^+ to flow spontaneously outward. This leakage flow progressively discharges the membrane, preventing an old stimulus from remaining stored indefinitely. If the stimulus ceases, $I_R(t)$ inevitably returns the neuron to rest at V_{rest} .

Substituting both physical definitions into the current balance at the membrane node:

$$I_{\text{syn}}(t) = C \frac{dV_m(t)}{dt} + \frac{V_m(t) - V_{\text{rest}}}{R}$$

This formulation balances currents (in Amperes), but to isolate the rate of change of potential $\frac{dV_m(t)}{dt}$, both sides are multiplied by resistance R :

$$R I_{\text{syn}}(t) = R \cdot C \frac{dV_m(t)}{dt} + [V_m(t) - V_{\text{rest}}]$$

We note that the product $R \cdot C$ possesses the physical dimensions of a time interval:

$$[R] \cdot [C] = \left(\frac{\text{V}}{\text{A}} \right) \cdot \left(\frac{\text{A} \cdot \text{s}}{\text{V}} \right) = \text{s} \quad (\text{or milliseconds, ms})$$

Defining this quantity as the **membrane time constant** $\tau_m = R \cdot C$, the equation is rearranged into the standard canonical form of the LIF model:

$$\tau_m \frac{dV_m(t)}{dt} = -[V_m(t) - V_{\text{rest}}] + R I_{\text{syn}}(t)$$

Computational Meaning: The final equation describes an ongoing compromise between two opposing forces:

1. **The external driving force ($R I_{\text{syn}}(t)$):** Attempts to push the neuron upward to reach the firing threshold V_{th} , converting input current into a voltage deflection scaled by resistance R .
2. **The elastic restoring force ($-[V_m(t) - V_{\text{rest}}]$):** Constantly acts to dissipate absorbed energy and drive the system back toward baseline rest.

Parameter τ_m quantifies the "useful memory window": a neuron with high τ_m retains charge longer and integrates temporally distant spikes; a neuron with low τ_m forgets past stimuli almost instantaneously, responding only to tightly synchronous events.

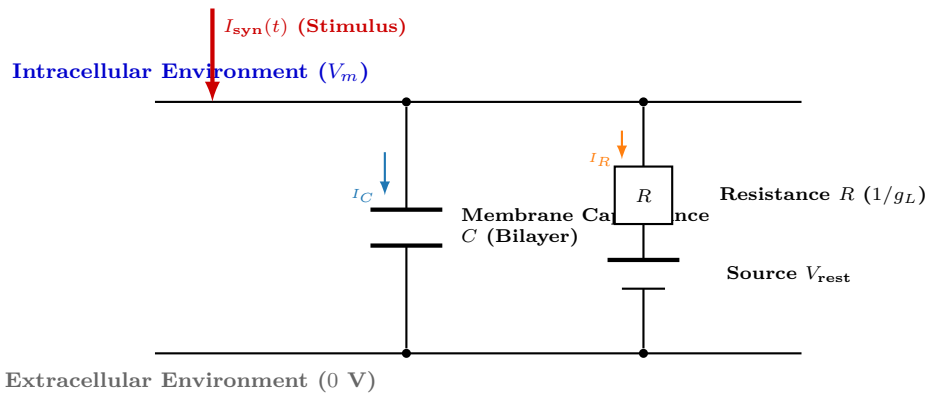


Figure 2: Equivalent RC membrane circuit of the LIF model. The lipid membrane acts as a capacitor C in parallel with passive ion channels modeled by leak resistor R in series with electromotive source V_{rest} .

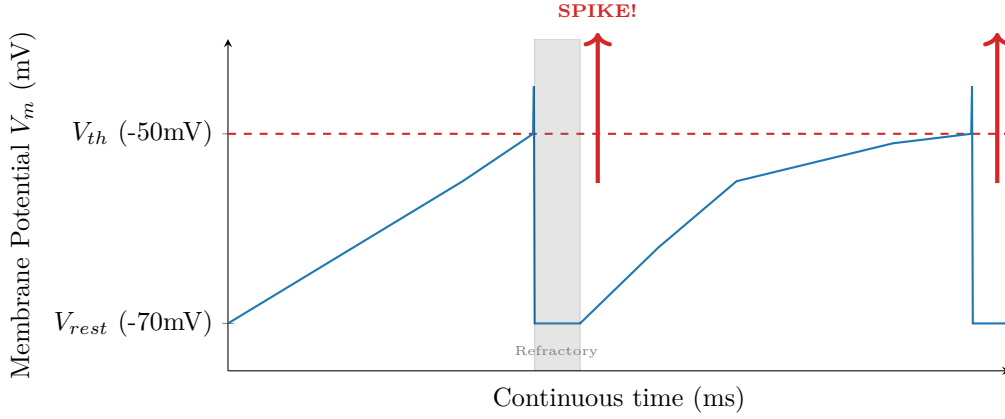


Figure 3: Membrane potential dynamics in a Leaky Integrate-and-Fire (LIF) neuron. Input synaptic current integrates the potential (blue line). When threshold V_{th} (dashed red line) is reached, an asynchronous spike is emitted, and potential is reset to V_{reset} . A brief refractory period follows (gray area) during which integration is suspended.

Final formula:

$$\tau_m \frac{dV_m(t)}{dt} = -[V_m(t) - V_{rest}] + R I_{syn}(t)$$

where:

- $V_m(t)$ represents the **instantaneous membrane potential** of the neuron (measured in millivolts, mV). It is the core internal state variable: it quantifies the transmembrane electrical potential difference between intracellular cytoplasm and extracellular fluid at time t .
- $\frac{dV_m(t)}{dt}$ denotes the **time derivative** of the potential, i.e., the instantaneous charging or discharging rate of the membrane.
- $\tau_m = R \cdot C$ is the **membrane time constant** (expressed in milliseconds, ms). It regulates the temporal inertia of the system: it modulates how quickly accumulated charge leaks away passively through ion channels.
- $-[V_m(t) - V_{rest}]$ constitutes the **passive leakage term**. It acts like an elastic restoring force: in the absence of external stimuli, it exponentially pulls the potential back toward the asymptotic resting equilibrium.
- V_{rest} is the **resting potential** (typically -70 mV), corresponding to the baseline energy level of the unperturbed membrane.
- R represents the **passive membrane resistance** (expressed in megaohms, $M\Omega$), quantifying opposition to ion transit across the phospholipid bilayer.
- $I_{syn}(t)$ is the **total incoming synaptic current** (expressed in nanoamperes, nA), derived from the weighted sum of incident presynaptic spikes.

Spike emission and subsequent reset occur according to the non-differentiable threshold rule (V_{th} is the threshold potential below which no spike is emitted):

$$\text{If } V_m(t) \geq V_{th} \implies \begin{cases} S(t) = 1 & \text{(Spike emission)} \\ V_m(t^+) = V_{reset} & \text{(Forced reset)} \end{cases}$$

Practical Example and Numerical Tracking: Consider a LIF neuron characterized by the following standard biophysical parameters:

- $V_{rest} = -70$ mV, $V_{th} = -50$ mV, $V_{reset} = -70$ mV; (simply the value to which V_m resets after firing)
- $\tau_m = 10$ ms, $R = 10$ $M\Omega$.

At initial time $t = 0$ ms, the neuron is at rest ($V_m(0) = -70$ mV). From this point onward, a constant synaptic current $I_{\text{syn}} = 2.5$ nA is applied.

Solving this step differential equation, the analytical trajectory takes the form:

$$V_m(t) = V_{\text{rest}} + R I_{\text{syn}} \left(1 - \exp \left(-\frac{t}{\tau_m} \right) \right)$$

Notice that the asymptotic steady-state potential is:

$$V_{\infty} = V_{\text{rest}} + R I_{\text{syn}} = -70 \text{ mV} + (10 \text{ M}\Omega \cdot 2.5 \text{ nA}) = -70 + 25 = -45 \text{ mV}$$

$V_{\infty} = V_{\text{rest}} + R I_{\text{syn}}$ represents the **asymptotic steady-state potential**: it constitutes the limit toward which membrane potential $V_m(t)$ would converge as $t \rightarrow \infty$ under constant current I_{syn} , in the absence of a threshold mechanism.

1. **At time $t = 5$ ms ($t = 0.5 \tau_m$):**

$$V_m(5) = -70 + 25 \cdot (1 - e^{-0.5}) = -70 + 25 \cdot (1 - 0.6065) = -70 + 9.84 = -60.16 \text{ mV}$$

The potential has climbed nearly 10 mV, but $V_m(5) < -50$ mV: the threshold is not crossed ($S(5) = 0$).

2. **At time $t = 10$ ms ($t = 1.0 \tau_m$):**

$$V_m(10) = -70 + 25 \cdot (1 - e^{-1.0}) = -70 + 25 \cdot (1 - 0.3679) = -70 + 15.80 = -54.20 \text{ mV}$$

Charge continues to accumulate, but passive leakage progressively slows the rate of increase ($S(10) = 0$).

3. **Exact instant of spike emission (t_{fire}):** Setting $V_m(t_{\text{fire}}) = V_{\text{th}} = -50$ mV:

$$-50 = -70 + 25 \cdot \left(1 - e^{-\frac{t_{\text{fire}}}{10}} \right) \implies 20 = 25 \cdot \left(1 - e^{-\frac{t_{\text{fire}}}{10}} \right)$$

$$0.80 = 1 - e^{-\frac{t_{\text{fire}}}{10}} \implies e^{-\frac{t_{\text{fire}}}{10}} = 0.20 \implies -\frac{t_{\text{fire}}}{10} = \ln(0.20) \approx -1.6094$$

$$t_{\text{fire}} = 10 \cdot 1.6094 \approx 16.09 \text{ ms}$$

At $t = 16.09$ ms, the potential reaches -50 mV: the neuron fires a binary pulse $S(16.09) = 1$, and its internal potential collapses instantly to reset value $V_m(16.09^+) = -70$ mV, clearing stored charge memory and restarting the integration cycle.

The spiking condition is defined discontinuously:

$$\text{If } V_m(t) \geq V_{th} \implies \begin{cases} \text{Emit Spike } S(t) = 1 \\ V_m(t^+) = V_{reset} \end{cases}$$

Following a spike, the neuron frequently enters a refractory period during which it cannot generate further pulses.

3.2 Advanced and Conductance-Based Models

For applications requiring greater physiological realism, more complex models are employed:

- **Hodgkin-Huxley Model (1952):** Describes ion channels (Sodium, Potassium) in detail using four coupled non-linear differential equations. It is the biophysical *gold standard*, but computationally prohibitive for large-scale networks.
- **Izhikevich Model:** Combines the rich dynamical repertoire of Hodgkin-Huxley (*bursting*, *chattering*, *adaptation*) with computational complexity close to LIF, using a system of two first-order differential equations.

4 Synapses and Synaptic Current Models

Neurons are not isolated; they communicate through synapses, which modulate the intensity of transmitted signals. In SNNs, synaptic weight w_{ij} determines the magnitude of the effect that a spike from presynaptic neuron i exerts on postsynaptic neuron j .

When a presynaptic spike arrives at time t_i^k , it generates a synaptic current $I_{syn}(t)$ in the postsynaptic neuron. Two primary formulations are used:

4.1 Delta-Current Model:

The spike produces an instantaneous current pulse modeled by a Dirac delta function. Membrane potential undergoes a discontinuous jump at the exact instant the spike arrives:

$$V_j(t^+) = V_j(t^-) + w_{ij}$$

where:

- $V_j(t^-)$ represents the **membrane potential of postsynaptic neuron j an infinitesimal instant before** the arrival of the presynaptic spike.
- $V_j(t^+)$ represents the **membrane potential of postsynaptic neuron j an infinitesimal instant after** pulse integration.
- w_{ij} is the scalar **synaptic weight** associated with the connection from presynaptic neuron i to postsynaptic neuron j ; it defines the magnitude of the instantaneous voltage step induced by the event.

This is the most computationally economical model and is widely adopted in deep learning SNN architectures.

4.2 Conductance-Based (Exponential) Model:

Unlike the delta-current model (in which potential undergoes an artificial step jump), this approach faithfully reproduces real biochemical interaction. The process unfolds via two distinct, coupled dynamics:

1. **Opening and Conductance Decay Phase (g_{syn}):** Upon arrival of a spike at time t_i^k , neurotransmitter molecules saturate postsynaptic receptors in fractions of a millisecond. Consequently, **membrane permeability (conductance g_{syn}) undergoes an instantaneous step** proportional to weight w_{ij} . Subsequently, ion channels close progressively, producing an **exponential decay** of conductance back toward zero.

This dynamic is governed by the first-order linear differential equation:

$$\tau_{syn} \frac{dg_{syn}(t)}{dt} = -g_{syn}(t) + w_{ij} \sum_k \delta(t - t_i^k)$$

where:

- $g_{syn}(t)$ represents the **instantaneous synaptic conductance** (measured in siemens, S, or nanosiemens, nS). It is the reciprocal of resistance ($g = 1/R$) and quantifies the overall open state of activated ion channels: zero indicates fully shut channels, whereas a high value indicates substantial opening to ion transit.
- $\frac{dg_{syn}(t)}{dt}$ denotes the **time derivative of conductance**, describing the instantaneous rate of protein pore opening or closing.
- τ_{syn} is the **synaptic time constant** (typically between 2 and 5 ms); it sets how quickly neurotransmitter is cleared and channels close.
- $-g_{syn}(t)$ constitutes the **spontaneous decay term**, responsible for the exponential relaxation of conductance toward the resting baseline.
- w_{ij} is **synaptic efficacy** (weight), modeling the impulsive step in conductance generated by each individual presynaptic event (proportional to the number of unblocked channels).
- $\sum_k \delta(t - t_i^k)$ represents the discrete **presynaptic spike train** emitted by afferent neuron i , modeled analytically as ideal Dirac pulses centered at times t_i^k .

2. **Current Injection Phase and Continuous Potential Evolution (V_j):** With ion channels temporarily open ($g_{\text{syn}}(t) > 0$), extracellular ions flow inward driven by the electrochemical gradient, generating an actual current $I_{\text{syn}}(t)$. This current charges membrane capacitance C , producing a **continuous, gradual rise in potential** (known as an excitatory postsynaptic potential, *EPSP*): the potential does not jump discontinuously, but traces a smooth curve reaching a rounded peak before returning toward rest as channels close.

The intensity of current flow obeys the generalized Ohm’s law for variable conductances:

$$I_{\text{syn}}(t) = g_{\text{syn}}(t) (V_j(t) - E_{\text{rev}})$$

where:

- $I_{\text{syn}}(t)$ is the **resulting instantaneous synaptic current** (expressed in nanoamperes, nA) entering the cytoplasm of postsynaptic neuron j .
- $V_j(t)$ is the **instantaneous membrane potential** of the receiving neuron.
- E_{rev} is the **reversal potential**: it defines whether the synapse’s effect is **excitatory** or **inhibitory**. It represents the limit value toward which synaptic current tries to pull the membrane potential:
 - **Excitatory Synapse** ($E_{\text{rev}} > V_{\text{th}}$, typically $\approx 0 \text{ mV}$): the current depolarizes the membrane, driving $V_j(t)$ upward toward firing threshold.
 - **Inhibitory Synapse** ($E_{\text{rev}} < V_{\text{th}}$, typically ≈ -70 – -80 mV): the current hyperpolarizes the membrane or holds it down, drawing $V_j(t)$ away from threshold and suppressing spike emission.
- $(V_j(t) - E_{\text{rev}})$ represents the **electrochemical driving force**, i.e., the effective potential difference pushing ions across the membrane through open channels.

NOTE: We do not need to track the absolute count of Na^+ ions inside the neuron, as they are merely the medium; what matters is the current intensity at any given time step.

5 Learning in Neuromorphic Networks

In conventional Artificial Neural Networks (ANNs), learning is a purely algebraic process: updating a matrix of continuous weights to minimize a global loss function. In Spiking Neural Networks (SNNs), learning must address both biological plausibility and severe mathematical challenges posed by the discontinuous nature of spikes.

5.1 Biological Foundations: What Does ”Learning” Mean Physically?

Before analyzing algorithms, it is crucial to clarify what a synaptic weight update Δw_{ij} represents physically in the brain. Biological learning, known as **synaptic plasticity**, is the ability of synapses to strengthen or weaken their efficacy over time.

Physically, a ”stronger” synapse (a larger w_{ij}) means a single incoming spike produces a larger postsynaptic potential (EPSP), driving the receiving neuron closer to its firing threshold. This increase in efficacy occurs via two main molecular mechanisms:

1. **Postsynaptic Modifications (Receptor Insertion):** When a synapse undergoes Long-Term Potentiation (**LTP**), the receiving cell physically inserts additional AMPA receptors (responsible for opening ion channels; each AMPA receptor has its pore) into its dendritic membrane. Upon subsequent glutamate release, more channels open simultaneously: conductance g_{syn} is higher, more Sodium (Na^+) ions enter, and potential V_m displays a much steeper rise.
2. **Presynaptic Modifications (Vesicular Release):** Potentiation can also prompt the presynaptic terminal to release a larger number of neurotransmitter vesicles for the same influx of Calcium (Ca^{2+}).

Conversely, if a synapse proves ineffective or counterproductive, receptors are internalized via endocytosis and removed from the membrane (Long-Term Depression, **LTD**): weight w_{ij} decreases and the synapse is silenced.

Learning algorithms for SNNs attempt to replicate or optimize this weight adaptation, falling into three main paradigms.

5.2 1. Unsupervised Learning: Hebbian Plasticity and STDP

This paradigm is rooted in biological local self-organization rules, encapsulated by Donald Hebb’s celebrated 1949 postulate: *“Neurons that fire together, wire together”*. In temporal SNNs, this rule takes the mathematical form of **Spike-Timing-Dependent Plasticity (STDP)**.

STDP establishes that firing together is not enough; causal temporal order is fundamental. Weight update Δw_{ij} depends strictly on the temporal offset $\Delta t = t_{\text{post}} - t_{\text{pre}}$ between the afferent neuron’s spike (t_{pre}) and the receiving neuron’s spike (t_{post}).

The Biological Coincidence Detector (NMDA Receptors): Biologically, STDP is mediated by NMDA receptors, which function as coincidence detectors. They trigger the insertion of additional AMPA receptors by letting in further Calcium (Ca^{2+}) **only when two conditions occur simultaneously**: glutamate binds (presynaptic signal) AND the membrane is already strongly depolarized (postsynaptic signal). This massive Calcium influx triggers the chemical cascade that inserts new AMPA receptors (LTP).

Molecular Calcium Dynamics and Dual Time-Scale LTP: To fully grasp synaptic plasticity, it is crucial not to confuse the two distinct roles Calcium (Ca^{2+}) plays on opposite sides of the synaptic cleft:

- **Presynaptic Calcium (Routine Transmission):** Enters the axon terminal through voltage-gated VGCC channels when an action potential arrives, acting as the exclusive mechanical trigger for **glutamate** exocytosis¹;
- **Postsynaptic Calcium (Learning Signal):** Enters the dendritic spine exclusively through NMDA channels when these are unblocked by pre-post temporal coincidence. In this context, the NMDA receptor acts only as a gating threshold: it is the intracellular Ca^{2+} influx that acts as a second messenger, activating enzymatic cascades that physically direct the insertion of new AMPA receptors.

¹Note the distinction with *glutamine*, which represents the inert form recycled by astrocytes and transported back to the neuron to be converted back into active neurotransmitter.

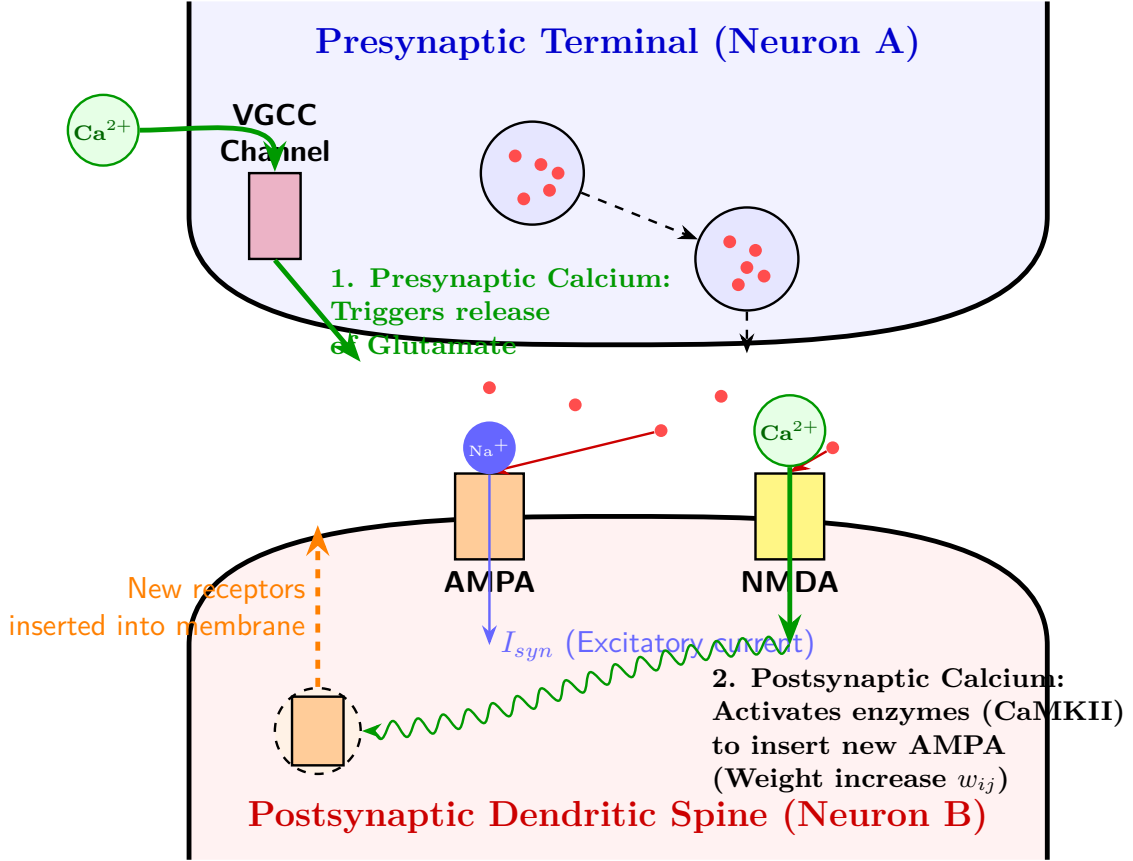


Figure 4: Diagram of the dual role of Calcium (Ca^{2+}) in the synapse. **Top (Presynaptic):** calcium enters through VGCC channels and acts as a mechanical trigger to release glutamate. **Bottom (Postsynaptic):** calcium enters through NMDA channels (only when unblocked) and acts as an enzymatic messenger directing the insertion of new AMPA receptors (synaptic plasticity, LTP).

The resulting physical reconfiguration of synaptic weight w_{ij} develops across two distinct time scales:

1. **Fast Phase (tens of seconds to a few minutes) — Reserve Recruitment:** The neuron does not need to synthesize new proteins immediately; Ca^{2+} -mediated enzymatic activation drives the rapid fusion of intracellular vesicles already pre-loaded with AMPA receptors. These are exposed and anchored to the postsynaptic density (PSD), instantly increasing conductance g_{syn} and thus the EPSP amplitude;
2. **Slow Phase and Consolidation (hours to days) — Gene Expression and Remodeling:** If stimulation and calcium influx persist regularly, the biochemical signal reaches the cell nucleus, activating transcription factors (such as CREB). The cell initiates *de novo* synthesis of protein complexes and morphological expansion of the dendritic spine, stabilizing the synaptic weight and converting a transient modification into long-term structural memory.

5.2.1 Mathematical Training Rule:

To translate this into a functional format for virtual simulation, the mathematical update rule for the plasticity window is:

$$\Delta w = \begin{cases} A_+ \exp\left(-\frac{t_{\text{post}} - t_{\text{pre}}}{\tau_+}\right) & \text{if } t_{\text{post}} > t_{\text{pre}} \quad (\text{LTP: Potentiation}) \\ -A_- \exp\left(-\frac{t_{\text{post}} - t_{\text{pre}}}{\tau_-}\right) & \text{if } t_{\text{post}} < t_{\text{pre}} \quad (\text{LTD: Depression}) \end{cases}$$

where A_+ , A_- determine the maximum amplitude of change and τ_+ , τ_- are the decay time constants (typically around 20 ms).

- **Causality (LTP):** If the presynaptic neuron fires *shortly before* the postsynaptic one ($t_{\text{pre}} < t_{\text{post}}$), it *physically contributed* to driving the voltage toward threshold. The synapse is rewarded and strengthened. The closer the two events, the sharper the increase.

- **Anti-Causality (LTD):** If the presynaptic neuron fires *after* the postsynaptic neuron has already emitted its spike, its input arrived late and was entirely useless. The network penalizes this inefficiency by depressing the synaptic weight, conserving energetic resources.

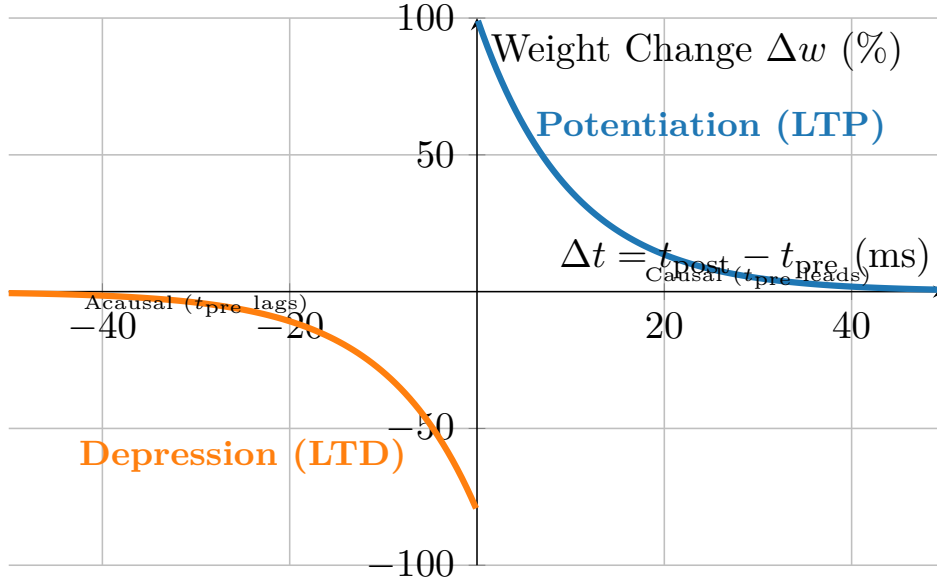


Figure 5: The Spike-Timing-Dependent Plasticity (STDP) window. If the presynaptic signal precedes the postsynaptic spike ($\Delta t > 0$, LTP), the synapse is exponentially strengthened based on temporal proximity. If the signal arrives late ($\Delta t < 0$, LTD), it is depressed. This asymmetric rule allows the network to autonomously extract causal temporal patterns without global supervisory labels.

5.3 2. Supervision via Surrogate Gradients

While STDP is excellent for unsupervised feature extraction, training deep neuromorphic networks on specific tasks (e.g., autonomous driving, complex classification) requires **Backpropagation**.

The Dead Neuron Problem (Vanishing Gradient): To apply the chain rule of differential calculus (BPTT - Backpropagation Through Time), we must compute the derivative of the neuron's activation function. The core obstacle in SNNs is that spike generation is governed by the discontinuous Heaviside step function:

$$S(t) = \Theta(V_m(t) - V_{th}) = \begin{cases} 1 & \text{if } V_m(t) \geq V_{th} \\ 0 & \text{otherwise} \end{cases}$$

The exact derivative of this discontinuous function is the Dirac delta $\delta(V_m - V_{th})$. This means the derivative is **zero everywhere** (the curve is flat), except at the exact threshold point where it is infinite. If we try to backpropagate the error to update weights ($\frac{\partial \text{Loss}}{\partial w}$), the gradient is multiplied by zero at each layer. The optimization halts immediately, and weights are never updated.

The Solution: The Surrogate Gradient: To circumvent this mathematical roadblock, a modified *straight-through estimator* (STE) approach is used. The idea decouples network behavior between forward and backward passes:

- **Forward Pass (Inference):** The exact, discontinuous biophysical equation is used. Neurons communicate exclusively via rigid binary spikes (0 or 1). Low-power neuromorphic properties are fully preserved.
- **Backward Pass (Training):** The problematic derivative of the Heaviside step is mathematically replaced with a smooth, differentiable surrogate derivative defined everywhere.

A common surrogate is the derivative of a scaled sigmoid (Fast Sigmoid) or a triangular function:

$$\frac{\partial S}{\partial V_m} \approx \sigma'(V_m) = \frac{1}{(1 + \alpha|V_m - V_{th}|)^2}$$

where α is a hyperparameter controlling gradient width (and thus tolerance) around the threshold. Thanks to this smooth surrogate, error gradients flow backward through time (BPTT) and across layers, updating synaptic weights w_{ij} . This technique has enabled SNN integration with modern deep learning frameworks (PyTorch, JAX), yielding accuracy competitive with ANNs.

Numerical Example: To illustrate how this substitution operates during optimization, consider a single neuron with threshold $V_{\text{th}} = -50\text{mV}$. Suppose that at a given instant its membrane potential is close to threshold but has not crossed it:

$$V_m = -50.2\text{mV}$$

Suppose further that, when backpropagating global error $\mathcal{L}$, the incoming gradient from subsequent layers is $\frac{\partial \mathcal{L}}{\partial S} = -2.0$ (loss derivative). Applying the chain rule to update an incoming weight w :

$$\frac{\partial \mathcal{L}}{\partial w} = \frac{\partial \mathcal{L}}{\partial S} \cdot \frac{\partial S}{\partial V_m} \cdot \frac{\partial V_m}{\partial w}$$

Let us analyze what happens across both passes:

1. **Forward Pass (Physical Reality):** The neuron evaluates the actual Heaviside step function:

$$S = \Theta(V_m - V_{\text{th}}) = \Theta(-50.2 - (-50.0)) = \Theta(-0.2) = 0$$

Because the threshold is not reached, the emitted spike is strictly **0**. Inference and forward dynamics remain binary and faithful to the spiking model.

2. **Backward Pass (Failure of the True Derivative):** If we used the true derivative of the Heaviside function ($\delta(x)$, zero everywhere except at threshold), we would obtain:

$$\left. \frac{\partial S}{\partial V_m} \right|_{\text{true}} = 0 \implies \frac{\partial \mathcal{L}}{\partial w} = (-2.0) \cdot 0 \cdot \frac{\partial V_m}{\partial w} = 0$$

Although the neuron was only 0.2mV away from firing, the error is nullified. The gradient vanishes, and weight w receives no update.

3. **Backward Pass (Surrogate Gradient Substitution):** In backpropagation, we replace $\frac{\partial S}{\partial V_m}$ with the derivative of a smooth surrogate function (such as a Fast Sigmoid with slope $\alpha = 1$):

$$\sigma'(V_m) = \frac{1}{(1 + \alpha|V_m - V_{\text{th}}|)^2}$$

Evaluating the surrogate at $V_m = -50.2\text{mV}$:

$$\left. \frac{\partial S}{\partial V_m} \right|_{\text{surrogate}} = \frac{1}{(1 + 1 \cdot |-0.2|)^2} = \frac{1}{(1.2)^2} = \frac{1}{1.44} \approx \mathbf{0.694}$$

Substituting this non-zero value into the chain rule:

$$\frac{\partial \mathcal{L}}{\partial w} \approx (-2.0) \cdot \mathbf{0.694} \cdot \frac{\partial V_m}{\partial w} = -1.388 \cdot \frac{\partial V_m}{\partial w} \neq 0$$

In summary: during the forward pass, the neuron remains a discrete event-based device (emitting 0), but during backward computation, the framework treats the neuron as having "almost fired", providing a useful gradient to adjust w in the proper direction.

5.4 3. ANN-to-SNN Conversion

An alternative approach to direct training with surrogate gradients is **ANN-to-SNN conversion**. Instead of optimizing the network within the discrete temporal space of spikes, an asymptotic equivalence is exploited: conventional continuous algebra can be replicated by a spiking network if information is encoded in the **average firing rate** (*Rate Coding*) over a sufficiently long time window.

Why Can Two Such Different Models Be Equivalent? At first glance, both paradigms seem incompatible: an ANN computes static continuous multiplications in a single step, while an SNN exchanges binary pulses distributed over time. The missing link is the behavior of the **Integrate-and-Fire (IF)** neuron without passive leakage (*non-leaky*), driven by an incoming spike train.

Consider the fundamental operation of the j -th neuron in an ANN with ReLU activation:

$$a_j^{\text{ANN}} = \max \left(0, \sum_i W_{ji} a_i^{\text{ANN}} + b_j \right)$$

In the IF neuron of its SNN counterpart, membrane potential $V_j(t)$ integrates binary input spikes $S_i(t) \in \{0, 1\}$ weighted by W_{ji} over time T :

$$V_j(T) = V_j(0) + \sum_{t=1}^T \left(\sum_i W_{ji} S_i(t) + I_{\text{bias}} \right) - V_{\text{th}} \sum_{t=1}^T S_j(t)$$

where the last term represents the reset mechanism (specifically, *reset-by-subtraction*, where emitting a spike subtracts exactly V_{th} from the potential rather than setting it to zero, preserving residual accumulated charge).

Dividing the entire equation by the total number of time steps T and defining the neuron's **average firing rate** as $r = \frac{1}{T} \sum_{t=1}^T S(t)$ (with $r \in [0, 1]$):

$$\frac{V_j(T) - V_j(0)}{T} = \sum_i W_{ji} r_i + \frac{I_{\text{bias}}}{T} - V_{\text{th}} r_j$$

If observation window T is sufficiently long ($T \rightarrow \infty$), the residual term on the left vanishes ($\frac{\Delta V}{T} \rightarrow 0$). Solving for output firing rate r_j :

$$r_j = \max \left(0, \frac{\sum_i W_{ji} r_i + b'_j}{V_{\text{th}}} \right)$$

Setting an appropriate scaling factor such that $V_{\text{th}} = 1$, the SNN firing rate r_j becomes **identical** to the continuous ReLU activation a_j^{ANN} . The SNN computes the same function not via a single scalar product of floating-point values, but by distributing the summation of pulses over T discrete steps.

NOTE: All neurons are evaluated over a fixed time window; once this window elapses, counts are converted back to a continuous value before proceeding to the subsequent layer.

The Conversion Pipeline:

1. **GPU Training (Standard ANN):** A standard neural network (e.g., ResNet or CNN) is trained on static data using only ReLU activations. It is critical to **remove bias terms** (or constrain them to zero) and avoid standard batch normalization during inference to prevent uncontrolled potential drift in spiking neurons.
2. **Weight Scaling and Threshold Normalization:** This is the critical step. In an ANN, a ReLU activation can theoretically assume arbitrarily large values (e.g., 14.5). In an SNN, the maximum physical firing rate is $r_{\text{max}} = 1$ (one spike per discrete timestep). If a continuous value exceeds this threshold, the SNN saturates (*clipping error*), losing information; conversely, if weights are too low, no neuron reaches threshold (*underflow error*). Weights or thresholds V_{th} for each layer are thus rescaled based on the maximum activation percentile observed over the training dataset:

$$W^{\text{SNN}} = W^{\text{ANN}} \cdot \frac{\lambda^{l-1}}{\lambda^l}$$

where λ^l is the maximum activation value found in layer l .

3. **Spiking Deployment:** Rescaled weights are loaded onto the SNN architecture. Continuous inputs (e.g., image pixels) are injected as constant current into the first layer or converted into spike trains via Poisson processes with rates proportional to pixel intensity.

The Fundamental Trade-Off: Accuracy vs. Latency The primary advantage of conversion is that it leverages the established Deep Learning ecosystem: deep architectures achieve virtually no accuracy loss ($< 1\%$) compared to their ANN counterparts on complex benchmarks like ImageNet.

The structural cost, however, is the **loss of temporal advantage**:

- To represent a continuous activation of 0.01 with high precision, the SNN must emit 1 spike every 100 time intervals. This necessitates a simulation window T on the order of hundreds or thousands of milliseconds ($T \approx 100\text{--}1000\text{ ms}$).
- This extensive inference latency multiplies total energy consumption (each timestep requires compute cycles on silicon) and forfeits instantaneous event-based processing (*Time-to-First-Spike*), rendering converted models uncompetitive for real-time reactive tasks aboard autonomous robotic platforms.

6 SNNs in Embodied AI: Advantages and Neuromorphic Hardware

Why invest in the complexity of SNNs when frame-based ANNs dominate? The answer lies in real-world operational efficiency.

- **Energy Efficiency (Cause-Effect Sparsity):** In ANNs, all neurons are computationally active on every frame. In SNNs, neurons consume energy **only when they fire a spike**. If the scene is static, the network remains silent. This extreme event-driven sparsity, when realized on native hardware, reduces energy consumption by orders of magnitude.
- **Temporal Latency and Reactivity:** SNNs process asynchronous information instantaneously. They do not wait to accumulate a "frame". This is critical for high-speed robotic control tasks (e.g., reactive manipulation, agile drones).
- **Neuromorphic Hardware:** Unlike massively parallel GPUs optimized for dense matrix multiplications, neuromorphic chips (Intel Loihi, IBM TrueNorth, SpiNNaker) are non-Von Neumann, massively distributed, asynchronous, event-driven architectures engineered specifically to emulate membrane integration and spike-based communication with minimal power.

7 Conclusions and Research Frontiers

Spiking Neural Networks represent a powerful, biologically grounded computational paradigm essential for the future of Embedded Artificial Intelligence and low-power autonomous systems. Although training remains challenging due to the non-differentiability of threshold mechanics, the emergence of surrogate gradient techniques and the maturation of neuromorphic hardware are bridging the performance gap with conventional ANNs.

Current research frontiers focus on merging supervised principles (surrogate gradients) with local unsupervised rules (STDP) for continuous on-chip learning, and developing increasingly efficient temporal encodings to unlock the true promise of zero latency and zero idle power in the absence of events.